Data Visualization Course Project

Spring 1405

Instructor

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Project Overview

The purpose of this project is to apply data visualization techniques to a real-world dataset and communicate insights effectively through graphical representations.

Students are expected to perform a complete data analysis workflow, including data preparation, exploratory data analysis, visualization, interpretation, and presentation of findings.

Project Requirements

Each group must select a real-world dataset from one of the following sources:

- Kaggle
- UCI Machine Learning Repository
- Open Government Data Portals
- World Bank Data
- Other approved public datasets

The dataset should contain at least:

- 500 observations
- 8 variables

Project Tasks

Task 1: Data Understanding

Provide a description of:

- Dataset source
- Dataset purpose
- Variables and their meanings
- Number of observations and features

Task 2: Data Cleaning

Perform necessary preprocessing steps such as:

- Missing value treatment
- Duplicate removal
- Data type correction
- Outlier investigation

Task 3: Exploratory Data Analysis (EDA)

Perform descriptive analysis and summarize key characteristics of the data.

Include:

- Summary statistics
- Variable distributions
- Relationships among variables

Task 4: Data Visualization

Create meaningful visualizations using Python libraries such as:

- Matplotlib
- Seaborn
- Plotly
- Altair

At least the following visualizations must be included:

- 2 Bar Charts
- 2 Line Charts

- 2 Distribution Plots
- 1 Correlation Visualization
- 1 Interactive Visualization

Additional visualizations are encouraged.

Task 5: Interpretation

For each visualization provide:

- Purpose of the figure
- Main findings
- Practical interpretation

Simply presenting plots without interpretation will not receive full credit.

Task 6: Infographic

Design a one-page infographic summarizing the most important findings of the project.

Deliverables

Each group must submit:

Dataset

Original dataset used in the project.

Python Notebook

A Jupyter Notebook containing all analyses and visualizations.

Final Report

A PDF report describing the methodology, results, and conclusions.

Infographic

A one-page infographic in PDF or PNG format.

Repository Structure

```
GroupName/
│
├── data/
├── notebook/
├── report/
├── infographic/
└── README.md
```

Evaluation Criteria

Item	Weight
Data Preparation	20%
Quality of Visualizations	30%
Interpretation of Results	25%
Report Quality	15%
Infographic Design	10%

Academic Integrity

All submitted work must be original. Any plagiarism or unauthorized use of external work will result in a zero grade.

Submission Deadline

To be announced by the instructor.